Rational Episodic Memory: When Should Wearable AI Remember?

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Abstract. Wearable AI systems rely on long-term memory to support retrieval, question answering, and contextual assistance. Yet an always-on system must decide what to preserve before knowing which experiences will later become relevant. We investigate two cognitively inspired principles for question-agnostic memory admission under fixed storage budgets. The first uses visible engagement as a rational-action-inspired proxy for relevance to the wearer's goals. The second uses surprise from pretrained visual representations as evidence of eventful change. We evaluate these signals as online write policies on long-form egocentric video, measuring both whether future-question-relevant intervals are substantially retained and how much of their duration is preserved. On eight long recordings averaging 4.14 hours, both policies outperform uniform sampling, while engagement achieves the strongest retention across all tested budgets. A secondary analysis on shorter, more event-dense recordings instead favors predictive surprise, consistent with prediction-error-based accounts of event segmentation. These results suggest that memory admission depends on the temporal structure of experience: surprise is effective for detecting densely occurring event changes, whereas engagement is particularly useful for preserving relevant episodes over extended everyday activity.

Keywords: Wearable AI · Egocentric Vision · Episodic Memory · Human-Centered AI

1 Introduction

Wearable AI systems promise persistent assistants capable of answering questions such as "Where did I leave my wallet?" or "When was the last time I took my medication?" Realizing this vision requires an always-on system that continuously observes the user's daily life. However, storing every observation is computationally impractical and quickly exceeds realistic memory budgets.

Humans face the same challenge. Although we receive a continuous stream of rich sensory information, we do not encode every detail into episodic memory. We rarely remember the color of every person's shoes or every crack in the sidewalk, even though those sensory signals were available to us. Instead, episodic memory is selective, preferentially encoding experiences that may be useful in the future.

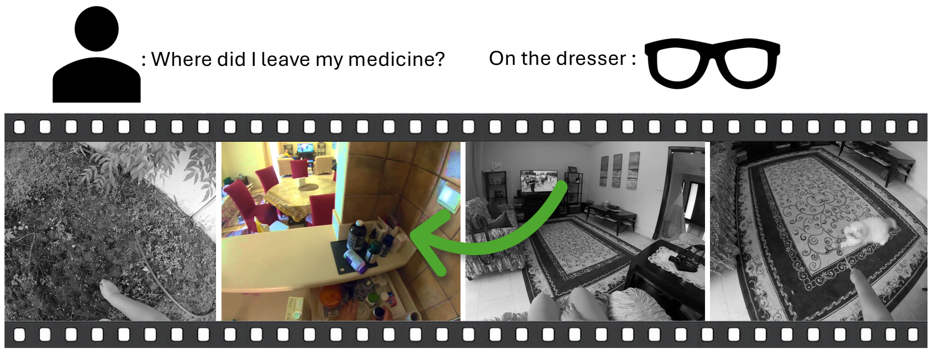

Fig. 1: A future user query requires localizing the relevant experience within a full day of egocentric observations.

We draw inspiration from this selectivity to study an upstream question in wearable AI memory: what should be written to memory in the first place? Prior wearable and egocentric systems construct memory through continuous capture, attention- or content-based selection, and subsequent compression or consolidation [1, 22, 23, 39, 42, 43]. However, comparatively little work treats memory admission as a distinct online problem: deciding which incoming experiences warrant long-term retention before the system knows what the user will later ask.

Cognitive Science offers many explanations for why an experience may be worth preserving. We draw on two well-established ideas that lend themselves to computational implementation: evidence of sustained goal-directed behavior and prediction error indicating eventful change.

Research on Theory of Mind and action understanding suggests that humans infer the goals of others by assuming that intelligent agents select actions rationally to achieve those goals [12]. Observed behavior can therefore provide evidence about what another person is trying to accomplish. An egocentric memory system similarly observes the wearer's behavior without direct access to their internal goals. This motivates visible engagement as a proxy for goal relevance: sustained interaction with an object, person, or environment suggests that the ongoing experience may be relevant to what the wearer is trying to accomplish.

Predictive-processing theories propose that perception continually compares incoming observations with expectations about what is likely to occur next [11], while Event Segmentation Theory argues that continuous experience is divided into discrete events when those expectations are violated [46]. This motivates our second candidate signal: change in the current visual representation relative to recent experience, used as a proxy for prediction-error-driven event boundaries. Our measure approximates the effect described by Itti and Baldi [19] and supported empirically by Kumar et al. [21] through distance in a pretrained latent space.

Motivated by these perspectives, we examine visible engagement and predictive surprise as alternative signals for question-agnostic memory admission. On long-form recordings, engagement retains more future-question-relevant evidence than surprise or uniform sampling, supporting the value of an agent-relative signal over extended experience. In a secondary analysis of shorter, more event-dense recordings, predictive surprise instead performs best. These recordings contain more frequent discrete actions and transitions, a setting in which prediction error may provide a particularly effective boundary signal, consistent with Event Segmentation Theory. Together, these results suggest that engagement is well suited to preserving sustained goal-relevant episodes, whereas surprise becomes more useful when experience is densely structured around distinct events.

Applications in wearable AI extend beyond episodic memory formation to deciding when a head-up display should surface context-aware information or when an AI assistant should proactively intervene.

Beyond wearable systems, similar principles may benefit long-duration autonomous robots operating under finite storage and compute. In particular, agent-relative signals grounded in the robot's own actions and task engagement could help determine which experiences warrant persistent memory during extended deployment.

These signals may also support large-scale egocentric and robotic dataset curation. Engagement-based selection could preferentially retain sustained goal-directed behavior, while surprise-based selection could emphasize event transitions and unusual changes.

2 Related Work

2.1 Egocentric Summarization and Memory

Retrospective Selection and Retrieval. Early egocentric summarization methods select representative frames or segments from an already recorded video using interaction, attention, coherence, diversity, or visual uniqueness [5, 24, 26, 30, 43]. Query-focused methods instead retrieve portions of an existing recording that match a supplied concept or question [31–34, 39]. These approaches improve summarization or retrieval after capture, but do not determine what should be stored before the user's future information needs are known.

Other systems compress or organize complete experiences into latent states, knowledge graphs, or user models [4, 38, 41]. Such representations can support efficient reasoning, but memory admission is typically learned indirectly from downstream supervision or replaced by continual structuring of the observed stream.

Online Memory Formation. Several systems make capture decisions as experience unfolds. StartleCam monitors skin conductivity and saves a buffer of recent images when it detects a startle response, while SenseCam captures images periodically and in response to changes detected by onboard light, motion, temperature, and passive-infrared sensors [17, 18]. Lin et al. classify the activity or scene of each incoming segment and apply a context-specific model to decide whether it should be retained in the summary, without first processing the complete recording [25]. AMEGO more directly constructs an active memory online by initiating object memories from hand-object interactions and identifying activity-centric locations [13]. These systems establish precedents for causal selection, but their admission criteria rely on additional sensors, highlight objectives, or predefined interaction and object cues.

Related egocentric work studies wearer-centered signals without necessarily applying them to memory formation. Su and Grauman infer engagement from egomotion, while other approaches use gaze or social engagement as evidence of wearer attention and subjective relevance [36, 37, 43]. These findings motivate our examination of engagement as a possible memory-admission signal, alongside predictive surprise.

2.2 Memory in Embodied and Language Agents

Embodied-agent memory systems commonly store observations in semantic maps, scene graphs, or timestamped databases for later retrieval and reasoning [2, 8, 14, 16]. Related work learns compressed latent memories through question-answering supervision or controls consolidation using repetition, similarity, and forgetting [4, 29].

Most closely related, *Worth Remembering* formulates admission as an explicit query-independent gating problem and stores episodes around moments of Bayesian surprise computed from changes in predictive visual representations [15]. Its results demonstrate that prediction error can outperform uniform and random storage for robot question answering. We examine this principle alongside visible engagement, asking whether unexpected change and sustained goal-directed behavior preserve different aspects of wearable experience.

Language-agent systems also perform explicit memory management through importance ranking, hierarchical storage, forgetting, linked-note evolution, or add/update/delete operations [7, 27, 28, 44, 48]. More recent approaches learn or specify admission using downstream answer quality, novelty, confidence, utility, or semantic surprise [35, 45, 47]. However, these methods generally decide whether already textualized facts or observations should be stored, rather than determining which portions of a continuous visual stream should first become memory candidates.

Our work instead studies the upstream perceptual decision of which egocentric observations should become memory candidates. We compare engagement and surprise as causal, query-independent write signals that can precede any particular representation, retrieval mechanism, or reasoning architecture.

3 Method

We evaluate two signals for determining which observations from a continuous egocentric video stream should be retained before future queries are known. First, a supervised engagement estimator predicts whether the wearer is behaviorally engaged with the current activity, providing a proxy for sustained goal relevance. Second, an unsupervised predictive-surprise measure estimates how strongly the current visual representation departs from recent experience, providing a signal of eventful change. We evaluate these signals independently as question-agnostic memory write policies under fixed storage budgets. The following sections describe their computation in detail.

3.1 Engagement

Definition and Model. We treat engagement as a supervised temporal classification problem: given an egocentric video window, predict whether it belongs to an annotated engagement episode, following the formulation of Su and Grauman [36]. Whereas their method proposes variable-length engagement intervals, we use fixed-length sliding windows to produce continuous estimates suitable for memory formation, where exact temporal boundaries are less important than retaining the relevant episode and limited surrounding context.

We represent each window using frozen V-JEPA2 features [3]. Because V-JEPA2 is pretrained to predict masked spatiotemporal representations, its embeddings capture motion, semantic continuity, and scene structure useful for engagement recognition. These features are provided to a Random Forest classifier trained on binary engagement labels using 64-frame windows with a stride of 32 frames.

The classifier contains 200 trees and uses random seed 0. At inference, it outputs the probability that each window belongs to an engagement episode. Predictions from overlapping windows are aggregated to produce a continuous engagement signal over time.

Annotations. We annotated a subset of the Wearable AI benchmark [40], which contains short egocentric videos spanning activities such as shopping, sightseeing, cooking, home improvement, exercise, and tourism. Our subset contains 40 videos totaling 7.05 hours, divided into 31 training videos and 9 held-out videos. All windows from a video remain within the same partition to prevent leakage.

Videos were selected to include both routine periods, such as walking or commuting, and sustained goal-directed activities, such as cooking or cleaning. Engagement intervals were annotated using the operational definition of Su and Grauman [36]: the wearer is engaged when they purposefully gather information about an object, scene, or ongoing task. Examples include inspecting products, reading signs, manipulating objects, or examining an unfamiliar environment. For activities not explicitly represented in the original dataset, we applied the same definition to the broader context. For example, an extended LEGO assembly sequence was treated as one continuous engagement episode because the wearer remained focused on locating pieces and constructing the model.

Engagement is inherently subjective, particularly at episode boundaries. Su and Grauman reported substantially higher agreement on whether engagement occurred (F1 = 0.914) than on its exact temporal boundaries (frame-level F1 = 0.818; interval-boundary F1 = 0.837) [36]. Although we did not conduct a formal multi-annotator consistency study, our objective is similarly to identify meaningful engagement episodes rather than recover frame-exact onset and offset times. Small boundary differences are therefore unlikely to materially affect the intended memory-write decision.

3.2 Surprise

Surprise is an unsupervised temporal novelty signal computed from the same V-JEPA 2 embeddings used for engagement estimation.

Let $z_t \in \mathbb{R}^d$ denote the pooled V-JEPA 2 embedding for a 64-frame window. Consecutive windows use a stride of 32 frames and therefore overlap by half their duration.

We define latent surprise as the standardized change between consecutive embeddings:

$$R_t = \frac{1}{2d} \sum_{i=1}^{d} \left(\frac{z_{t,i} - z_{t-1,i}}{\max(\sigma_i, \epsilon)} \right)^2,$$

where σ_i is estimated from a development set and $\epsilon > 0$ prevents division by zero. This is a dimension-normalized half squared Mahalanobis distance under a shared diagonal covariance model. We use it as a Bayesian-surprise-inspired proxy for latent visual change, rather than an exact Bayesian surprise measure.

We map R_t to $[0,1]$ using fixed calibration values estimated on development data disjoint from the evaluation set:

$$\widetilde{S}_t = \mathrm{clip}\left(\frac{R_t - r_{\min}}{r_{\max} - r_{\min}}, 0, 1 \right),$$

where $r_{\min} = 0.01$ and $r_{\max} = 0.25$. Finally, we apply causal exponential smoothing:

$$S_t = 0.25\widetilde{S}_t + 0.75 S_{t-1}, \qquad S_1 = \widetilde{S}_1.$$

The resulting $S_t \in [0,1]$ is held constant until the next 32-frame window is available.

4 Experiments

We evaluate engagement and surprise as question-agnostic memory-admission signals on 16 Ego4D videos totaling 42.71 hours. The evaluation includes a long-form regime and an analysis of shorter recordings. We compare both signals with uniform sampling under matched storage budgets. We additionally conduct a small comparison of V-JEPA and optical-flow representations for engagement estimation.

4.1 Memory Write Policy Evaluation

Setup. The evaluation contains eight shorter recordings averaging 1.20 hours and eight long recordings averaging 4.14 hours. Because our central research question concerns selective memory over extended experience, the long-video subset serves as the primary evaluation regime, while the shorter recordings provide a secondary comparison across temporal scales.

Although the subsets are defined by duration, they also differ qualitatively in temporal structure. The shorter recordings are generally concentrated around one sustained task or a closely related sequence of tasks, including baking, carpentry, gardening, cleaning, and exercise. The longer recordings instead span broader portions of everyday experience, combining multiple activities and transitions. These include outings that move between walking, conversation, sightseeing, restaurants, phone use, and transportation, as well as household recordings containing yard work, sewing, cleaning, and social interaction. We therefore report the two subsets separately while treating the long recordings as the setting most directly aligned with the long-horizon memory problem.

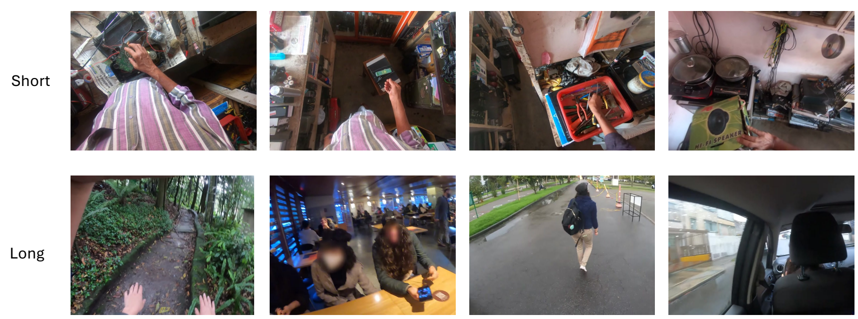

Fig. 2: Qualitative comparison across temporal regimes. Top: a shorter, task-focused recording in which engagement remains consistently high, while visual surprise better distinguishes transitions between subactions. Bottom: a long, heterogeneous recording in which engagement remains discriminative across routine periods and sustained goal-directed activity.

To evaluate whether a write policy retains useful information under a limited storage budget, we use the temporal intervals from Ego4D Natural Language Queries (NLQ) as targets. NLQ pairs natural-language questions about past egocentric experience with the intervals containing the visual evidence needed to answer them. Retaining these intervals therefore provides a proxy for whether the memory contains information that could support plausible future queries while discarding portions of the stream less relevant to those information needs.

Of the 16 selected videos, 10 contain official NLQ annotations, but only two of the eight long recordings do. For the remaining six long recordings, we construct 149 narration-derived annotations following the same question-and-temporal-interval format. We select concrete, question-worthy narrated events distributed throughout each recording, convert them into natural-language questions, and assign the intervals containing the evidence needed to answer them. Selection is performed without access to engagement or surprise scores, and results using official and researcher-created annotations are reported separately.

Each policy assigns scores to incoming temporal windows without access to future queries or evaluation annotations. At each storage budget, the highest-scoring windows are retained. Engagement and surprise are compared with uniformly spaced fixed-length windows selected to retain the same amount of video.

Results. We report results separately for the long and short recording subsets. For each matched storage budget, the tables report Hit@50, which counts an annotated interval as retained only when at least half of its duration is preserved, and annotated-time recall, which measures the fraction of total annotated duration retained. These metrics capture both the number of substantially preserved intervals and the temporal extent of relevant evidence maintained in memory.

Table 1: Memory retention across temporal regimes. Long videos contain 8 recordings averaging 4.14 hours with 215 intervals; short videos contain 8 recordings averaging 1.20 hours with 292 intervals. Entries report Hit@50 / annotated-time recall, in percentages.

	Long videos			Short videos		
Budget	**Engagement**	**Surprise**	**Uniform**	**Engagement**	**Surprise**	**Uniform**
30%	**42.3** / **44.6**	33.0 / 41.5	26.4 / 29.7	15.8 / 15.2	**28.8** / **31.6**	22.1 / 30.1
25%	**33.5** / **38.1**	28.8 / 35.9	22.4 / 25.2	12.0 / 13.0	**24.0** / **27.8**	17.6 / 25.0
20%	**30.7** / **32.6**	20.5 / 29.6	18.6 / 20.2	7.9 / 8.8	**17.5** / **23.3**	14.5 / 19.9
15%	**25.6** / **27.5**	13.5 / 21.5	13.0 / 15.0	7.2 / 7.2	**11.0** / **15.9**	10.9 / 15.0
10%	**16.7** / **18.1**	9.3 / 14.0	9.0 / 10.0	3.8 / 5.2	6.2 / **10.9**	**7.3** / 10.0
5%	**10.7** / **11.1**	4.2 / 7.3	4.2 / 4.8	2.7 / 3.2	2.4 / **5.4**	**3.5** / 4.9

The two temporal regimes produce a clear reversal in relative performance. On long recordings, engagement achieves the highest Hit@50 and annotated-time recall at every tested budget, with the largest gains appearing at tighter storage constraints. On shorter recordings, predictive surprise performs best across most budgets, while uniform sampling remains competitive only at the lowest budgets. These results indicate that the relative effectiveness of the two signals depends on the temporal structure of the evaluated video.

Table 2: Sensitivity to annotation source. Values report the engagement-minus-surprise gap in percentage points; positive values favor engagement. Official results use two long videos with Ego4D NLQ annotations, while researcher results use the remaining six long videos.

Budget	Official Hit@50	Researcher Hit@50	Official Time	Researcher Time
30%	+25.8	+8.6	+5.3	−0.9
25%	+10.6	+7.9	+3.8	−0.5
20%	+19.7	+11.5	+2.8	+1.1
15%	+22.7	+10.8	+6.5	+2.5
10%	+16.7	+5.8	+4.3	+1.5
5%	+12.1	+5.8	+3.9	+1.1

Annotation Consistency Check. The central Hit@50 result is consistent across annotation sources: engagement outperforms surprise at all six storage budgets for both the official and researcher-created intervals. The ordering of the gap across budgets is only moderately similar ($\rho = 0.580$), and the mean absolute difference in gap magnitude is 9.54 percentage points, so the size of the advantage should not be treated as stable. Annotated-time recall is less consistent: the two sources agree at four of six budgets, while the researcher-created intervals weakly favor surprise at 25% and 30% by only 0.5–0.9 percentage points. Given that the official comparison contains only two videos, these small reversals may reflect differences in video composition or sampling variability rather than a systematic annotation-source effect. This analysis therefore supports the robustness of the Hit@50 conclusion while motivating a larger independently annotated evaluation.

4.2 Engagement Representation Ablation

We additionally compare V-JEPA features with a motion-based engagement representation following Su and Grauman [36]. Dense Farnebäck optical flow is computed between successive frames using OpenCV [6,10], temporally smoothed, and summarized with a hierarchical spatiotemporal pyramid that pools horizontal and vertical motion statistics across temporal segments and spatial cells. Both representations use the same engagement annotations and training protocol described in Section 3. Among the configurations evaluated on the held-out videos, the highest-F1 settings were 128-frame windows with stride 64 for optical flow and 64-frame windows with stride 32 for V-JEPA.

Table 3: Engagement representation ablation. Results report the best evaluated window-size/stride configuration for each representation.

Representation	Win./Stride	F1	Prec.	Recall	Spec.	AUC
Optical Flow	128/64	0.883	**0.889**	0.877	**0.839**	0.914
V-JEPA	64/32	**0.904**	0.882	**0.927**	0.816	**0.951**

V-JEPA achieved higher F1, recall, and ROC AUC, while optical flow achieved slightly higher precision and specificity. The relatively small performance gap indicates that much of visible engagement can be captured from motion structure alone, consistent with Su and Grauman's use of long-term egomotion cues for engagement detection [36]. V-JEPA nevertheless detects a broader set of engaged moments, suggesting that its semantic representation captures additional cues beyond motion, while optical flow produces fewer false positives.

5 Discussion

5.1 Interpretation

The observed reversal across temporal regimes is broadly consistent with the intended roles of the two signals. Engagement asks whether the wearer appears to be acting purposefully in pursuit of an ongoing goal. This signal is most informative in long, varied recordings that contain a mixture of routine periods and distinct activities, such as walking, driving, eating, working, and interacting with other people. In this setting, engagement can distinguish relatively sparse periods of sustained goal-directed behavior from the broader background of everyday experience, which may explain its stronger performance on the long-form videos.

The shorter recordings are qualitatively different. They are more often organized around a single sustained activity, such as cooking, cleaning, gardening, or exercise, with relatively little inactive or unrelated footage. We hypothesize that the engagement signal becomes saturated in this setting: when the wearer appears engaged throughout most of the recording, engagement provides little discrimination about which moments should receive a limited memory budget. This explanation remains preliminary until the prevalence and temporal distribution of predicted engagement are measured directly across the two subsets.

In these denser, task-focused recordings, predictive surprise becomes more useful. Even when the wearer remains engaged throughout an activity, the stream still contains changes in objects, subactions, locations, and task state. A signal based on representational change can therefore identify transitions within the larger activity, consistent with the prediction-error mechanism emphasized by Event Segmentation Theory. Taken together, the results suggest that engagement and surprise are not competing estimates of a single notion of importance. Engagement is most useful when goal-directed behavior is sparse relative to the full stream, whereas surprise becomes more useful when engagement is widespread and memory formation instead depends on segmenting a dense sequence of events.

Long, task-saturated case. To separate recording duration from activity structure, we additionally analyze a multi-hour welding recording dominated by one sustained, highly engaged task. Because this recording differs qualitatively from the varied long-form videos, it is excluded from the primary aggregate and treated as a separate case study.

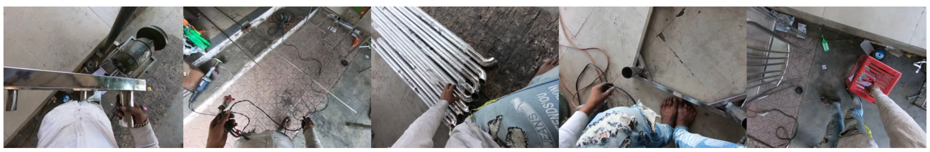

Fig. 3: Qualitative samples from the long, task-saturated welding recording. Although the recording spans several hours, the wearer remains engaged in a single sustained activity. Engagement is therefore broadly elevated, while surprise provides greater discrimination among changes in tools, materials, viewpoints, and welding subactions.

Table 4: Memory retention on a long, task-saturated welding recording. The video is dominated by one sustained activity. Entries are percentages.

	Hit@50		Annotated Time	
Budget	**Engagement**	**Surprise**	**Engagement**	**Surprise**
30%	26.3	**61.4**	42.1	**52.2**
25%	24.6	**42.1**	**41.5**	40.8
20%	17.5	**29.8**	**34.8**	30.1
15%	15.8	**19.3**	**29.4**	22.2
10%	**14.0**	3.5	**22.8**	10.8
5%	**7.0**	1.8	**10.6**	0.3

The welding case produces a budget-dependent result. Surprise achieves substantially higher Hit@50 at moderate and high storage budgets, despite the recording being long, while engagement becomes stronger under the tightest budgets and preserves more annotated duration at most operating points. This provides evidence against duration alone explaining the difference between the two main subsets. Instead, the temporal structure of the recording appears important: when one sustained task causes engagement to remain broadly elevated, surprise can provide greater discrimination by identifying changes within that activity. As a single-video case study, however, this result should be interpreted as supporting evidence rather than a general conclusion.

5.2 Limitations

This work is an early exploration conducted on a limited number of recordings and does not yet evaluate complete day-long wearable deployments. The study also depends on researcher-created annotations. Engagement intervals were labeled by a single annotator using the operational definition of Su and Grauman [36].

For selected Ego4D recordings without official NLQ annotations, we additionally constructed narration-derived questions and temporal intervals representing information that a wearer might reasonably ask about later. These intervals were selected without access to engagement or surprise scores.

Both the learned engagement model and the retention evaluation may therefore reflect subjective annotation decisions.

The proposed explanation for the difference between short and long recordings also remains preliminary. We hypothesize that engagement becomes saturated in shorter, task-focused recordings, while surprise benefits from their denser sequence of event transitions. Because duration and activity composition vary together in the present evaluation, we cannot truly isolate which factor causes the observed reversal.

5.3 Future Work

Future evaluation should use larger-scale day-long recordings, independent multi-annotator labeling, and datasets that disentangle recording duration from activity composition.

Beyond Visual Memory. While this work estimates engagement primarily from visual behavior, future systems may benefit from multimodal sensor fusion to better infer latent user intent. Early wearable memory systems explored physiological signals such as galvanic skin response and electroencephalography to identify memorable experiences or subjective interest [1, 17]. More recent wearable platforms instead provide complementary behavioral signals, including eye tracking, inertial measurements, and egocentric vision, enabling richer estimation of user attention and interaction. For example, gaze has been shown to improve egocentric video summarization by providing a direct measure of visual attention [43], while platforms such as Project Aria integrate synchronized cameras, eye tracking, IMUs, microphones, and additional sensors into a unified wearable research platform [9]. Recent work has also demonstrated user-generalizable surface electromyography (sEMG) for wearable human-computer interaction, suggesting that muscle activity could provide an additional cue for manipulation intent in future smart glasses [20]. We hypothesize that such signals are most effective when fused with complementary modalities rather than used in isolation, allowing ambiguities to be resolved—for example, distinguishing intentional object manipulation from unrelated muscle activations such as scratching one's arm.

Toward User Models. While the present work approximates surprise using local semantic feature dynamics, a promising direction is to compute surprise relative to a persistent personalized user model. Rather than modeling only recent observations, such a model could encode stable knowledge about recurring environments, objects, routines, and individuals, allowing surprise to reflect violations of long-term expectations. Recent work has begun constructing persistent user models from long-term egocentric interactions to capture behavioral regularities and preferences [41]. Likewise, Gorlo et al. note that local surprise models can repeatedly trigger on recurring observations and identify habituation as an important direction for future work [15]. Integrating Bayesian surprise with a richer personalized user model may therefore enable more human-like memory formation, where novel observations are initially encoded but gradually become unsurprising as the internal model adapts.

6 Conclusion

We present preliminary evidence that visible engagement and predictive surprise provide useful, cognitively motivated signals for question-agnostic memory admission in wearable AI. Their effectiveness depends on the temporal structure of experience. Predictive surprise performs well in shorter, event-dense recordings, where changes between actions and task states provide informative boundaries consistent with Event Segmentation Theory. Engagement is more effective in long, heterogeneous recordings, where sustained goal-directed behavior helps distinguish potentially useful episodes from routine or passive portions of everyday experience.

These results suggest that no single signal captures every reason an experience may be worth preserving. Surprise provides a mechanism for identifying eventful change, while engagement grounds memory formation in the wearer's own behavior and goals. Although the present evaluation remains limited in scale, it offers early evidence that cognitively motivated write policies can help the wearable AI community design selective memories suited to different deployment settings and storage constraints.

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